G H Raisoni International Skill Tech University, Pune.

Question Paper

Continuous Assessment Examination CAE – II (CAE-II)

Programme: B Tech **Year: First** **Semester: I**

Maximum Marks: 20 Duration: 60 mins.

Course Name: Introduction to Cyber Security Course Code: OT109

Instructions:

1. Support your answers with examples where applicable. **Each question carries 5 marks.**
2. Students must label their answers clearly with the corresponding question number, e.g., Q.No. 1A, Q.No. 2A(i), etc.

Note: This question paper addresses below Course Outcomes (CO's):

CO1	Identify and differentiate various cyber threats and vulnerabilities.
CO2	Develop and recommend strategies to mitigate cybersecurity risks.

Q. No.	Sub Q. No.	Question (Any Two).
1		
	A	i) Explain the three types of firewalls: packet-filtering, stateful inspection, and next-generation firewalls (NGFW). Give one advantage of each type. ii) You are the IT manager of a small online retail store. The store website accepts customer payments and stores basic customer information. Which type of firewall would you recommend and why?
	B	i) Describe the difference between Network-Based IDS (NIDS) and Host-Based IDS (HIDS) ii) XYZ company's employee reports their laptop is running unusually slow, and you notice high network traffic from that device at odd hours. How would HIDS help in investigating this situation?
	C	i) Define Role-Based Access Control (RBAC) and give two advantages of using it. ii) Sourav manages IT for a hospital where doctors need access to patient records, nurses need access to daily care information, and administrative staff need access to billing information. Explain how you would organize these access rights using RBAC.
	D	i) Compare symmetric and asymmetric encryption. Include one example algorithm for each type. ii) Suman setting up a new office network where employees need internet access, but you also have an internal server with sensitive client data. How would you combine encryption and network segmentation to ensure the security of sensitive data?
2		
	A	i) Using the 2019 Business Email Compromise Attack case study, analyse how AI-powered phishing attacks leverage NLP to create convincing fake communications. What specific AI techniques were used? ii) Propose defensive strategies against AI-generated phishing attacks, incorporating both technical controls and user awareness training.
	B	i) Analyse the 2021 Deepfake Authentication Breach case study where attackers stole $35 million. What specific AI technologies were used, and how could this attack have been prevented. ii) Design a robust authentication system that can detect and prevent AI-generated deepfake attacks in real-time
	C	i) List three key steps for using AI-based vulnerability scanning tools to secure a company website. ii) Your colleague's computer is behaving differently after downloading a health advisory file. What immediate steps should be taken?
	D	i) How did Darktrace's AI system help protect against the WannaCry ransomware attack? ii) List two early warning signs that AI-powered threat detection could identify to prevent ransomware spread.